

3. API Design

The frontend communicates with the backend through REST API endpoints.

Generate Route

Endpoint:

POST /route/generate

Purpose:

Generate a route between the start point and the destination.

Request body:

start_point
goal_point
obstacles
drone_parameters

Response:

route_coordinates
distance

Return Home

Endpoint:

POST /mission/return-home

Purpose:

Generate a route from the current drone position to the original start point.

Land Drone

Endpoint:

POST /mission/land

Purpose:

Stop the current mission and mark the drone as landed.

Reset Mission

Endpoint:

POST /mission/reset

Purpose:

Reset mission state and clear route data.